



TECHNICAL SPECIFICATIONS FOR SWITCHING SERVICES <Apply for Member Organization>

PART III: RECONCILIATION FILE FORMAT

NATIONAL PAYMENT CORPORATION OF VIETNAM

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REVISION HISTORY

Version	Release Date	Editor	Reviewer	Description For Change
0.9	From 01/2016 - 11/2016	Ha Nam Ninh Nguyen Hung Cuong Bui Thi Kim Dung Dao Thanh Son Huynh Cong Linh Pham Minh Ngoc Le Anh Tuan Nguyen Thanh Quynh	Nguyen Hung Nguyen	- Built draft Technical Specification Document - Spitted the document to 05 parts include: • Part I – Transaction Processing • Part II – Message Format • Part III – Reconciliation File Format • Part IV – Security and Communication • Part V – Annex - Added international transaction processing on following part: • Part VI – Annex: International Transaction Processing
1.0	12/2016	Ha Nam Ninh Nguyen Hung Cuong Nguyen Thanh Quynh Dao Thanh Son Huynh Cong Linh	Nguyen Hung Nguyen	- Added some data elements on CHIP transaction at: • Part II – Message Format: Section 6 – Data elements, Section 7 –

				Message format - Part V – Annexa: Section 4 – Response code - Updated the Online switching, Tokenization process flow synchronized with regulations for member organizations participating in the NAPAS system at: - Part I – Transaction processing, Section 4.2, 4.3, 4.5 - Part II – Message format, Section 6.2 - Added service and interchange fees between NAPAS and Member Banks; added two fields : DE 22, DE 25 at : - Part III – Message structure : section 4.3
1.0	05/2017	Ha Nam Ninh Nguyen Khanh Quan	Nguyen Hung Nguyen	- Add the Reserve field information in the detail files to ensure there is a backup later. - It also adds a description of situation using Reserve Field for ECOM service.
1.0	07/2017	Ha Nam Ninh	Nguyen Hung Nguyen	- Updated Type and

		Nguyen Viet Trung		Description to ensure that all fields filled with align right and padding left with zero or space if their type is number or character respectively.
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1 Audience

“Technical Specifications on NAPAS System Connectivity” document is intended for the Member Organizations (MO) of NAPAS include but not limited to Member Banks of NAPAS and Payment Intermediaries that directly connected to card switching services.

2 Scope

The document describes the regulations about the reconciliation/ settlement file formats between NAPAS and MO while implement the switching services.

The switching services include but not limited to the following services:

2.1 Domestic card switching services

- a. Domestic card switching services at ATM: allow customers to use Member Banks card to perform transactions on the ATM equipment of the Member Banks in the Switching Organizations of other countries, and vice versa. Domestic card switching services on the ATM include but not limited to the following transactions:
 - Balance Inquiry
 - Cash Withdrawal
 - InTra-bank Fund Transfer (ITFT)
 - Mini Statement
 - PIN Change
- b. Domestic card switching service at POS: allow customers to use Member Banks card to perform transactions at POS of Member Bank of NAPAS network or Member Bank of Switching Organizations of other countries, and vice versa. Card switching services at POS include but not limited to the following transactions:
 - Purchase
 - Void
 - Balance Inquiry
 - PIN Change

- c. Online switching service (Ecom): allow customers to make purchase through the online payment website that is connected to NAPAS. A successful payment transaction consists of 03 messages after sending to Member Bank:

- Verify Cardholder information (Verify Card)
- Verify OTP
- Ecom Purchase

Beside normally online payment service, NAPAS also debits on customer account by processing only one message:

- Ecom Fast Purchase

- d. Inter-Bank Fund Transfer 24/7 service (IBFT): allows the customers to transfer money on many different payment channels (ATM, Internet banking, Mobile banking, Bank counters ...). In addition, this is also a service that Payment Intermediaries can use to allow a fund transfer from e-wallet account of Payment Intermediaries to account of Member Bank of NAPAS. This services include following messages:

- Beneficiary cardholder / account information inquiry (IBFT Inquiry)
- Beneficiary cardholder / account fund transfer (IBFT Deposit)

Note: *IBFT Inquiry message and IBFT Deposit message are 02 independent messages and NAPAS system does not check the relationship between these messages.*

- e. Payment code service: allows customer to use Member Bank payment channels which transfers and receives using Payment code (PC). Customer who transfers money is provided a Payment code when perform transfer transaction, beneficiary customer use this Payment code on channel of Bank to receive money. On-us transfer or off-us transfer bank and receive bank may be on the same bank or two different banks. The beneficiary customer object of service mainly direct to customers who have not got bank account.

- Payment code create request (PC Create)
- Payment code authentication (PC Verify)
- Payment code withdrawal (PC Cash Withdrawal)
- Payment code inquiry (PC Inquiry)

- f. Tokenization service: the service that encrypt customer information to Token ID range – a range codes to identify alternative using of customer information when conducting payment.
- Card information verify (Token Verify Card)
 - OTP verify (Token Verify OTP)
 - Token Purchase
 - Token Fast Purchase

2.2 Cross-border card switching services

- a. Cross-border card switching services at ATM/POS: allows customer to use the card of NAPAS Member Bank to perform transactions on the Member Banks ATM/POS equipment of the Switching Organization of other countries, and vice versa. Cross-border card switching services at ATM/POS include but not limited to the following transactions:
- Balance Inquiry at ATM
 - Cash Withdrawal at ATM
 - Balance Inquiry at POS
 - Purchase at POS
- b. Cross-Border Fund Transfer – CBFT: allows customers to use their bank cards in Switching Organizations of other countries can transfer money of account number/client card number of Member bank in NAPAS. CBFT service deploy to Member Bank based on the platform of the IBFT service, include two transactions:
- Beneficiary cardholder / account information inquiry (CBFT Inquiry)
 - Beneficiary cardholder / account fund transfer (CBFT Deposit)

3 Terms and Acronyms

The following table describes the terminology and acronyms used in this document:

No.	Terms/Acronyms	Description
1	SW	Switching system (Switch) of NAPAS
2	ACQ	Acquirer (Settlement Bank)
3	ISS	Issuer
4	BNB	Beneficiary Bank
5	PMI	Payment Intermediary
6	MO	Member organization (Include Member Bank và Payments Intermediary)
7	Bank MO	Bank Member Organization
8	ICA	International Card Association (include Visa, MasterCard, Amex, JCB, etc.)
9	ATM	Automated Teller Machine
10	POS	Point Of Sale/Service
11	ITFT	InTra-bank Fund Transfer (Transfer to internal bank)
12	IBFT	Inter-Bank Fund Transfer 24/7
13	CBFT	Cross-Border Fund Transfer
14	OTP	One Time Password
15	APN	Asian Payment Network
16	MEPS	Malaysia Electronic Payment System
17	KTFC	Korea Financial Telecommunications & Clearings Institue
18	NETS	Network For Electronic Transfer

19	ITMX	Interbank Transaction Management and Exchange
20	CUP	China Union Pay
21	NCC UC	Union Card

4 Reconciliation file

4.1 Reconciliation file structure

All transactions of card switching service between MO and NAPAS can be aggregated in a detail transactions file or separated into multiple files to perform the reconciliations between NAPAS and MO.

The format of the transactions file name are described as following:

File name format	Description
MMDDYY_YYY_BBB_999999_X_TC_<ServiceCode>.dat	The transaction detail file which contains the successful transactions in a session is created by Napas and sent to MO.
MMDDYY_YYY_BBB_999999_X_XL_<ServiceCode>.dat	The transaction detail file which contains the processed transactions in a session is created by Napas and sent to MO.
MMDDYY_YYY_BBB_999999_X_SL_<ServiceCode>.dat	The dispute transactions detail file is created by MO and sent to Napas.

Explains words and phrases in the file name:

Phrase	Description
MMDDYY	The creation date of file, in Month/Day/Year format. Noted : - MMDD filled with the value of DE#15 of message online which is set up by NAPAS system and YY is the last two numbers of the current year. - Daily with calendar day, NAPAS creates the reconciliation file, but to the nearest working day, NAPAS will send these reconciliation files to MO.

ZZZ	ZZZ corresponds to following codes : ISS : Bank is Issuer ACQ : Bank is Acquirer BNB : Bank is Beneficiary (IBFT). Noted : with IBFT transaction, when MO play roles of ACQ=ISS then transaction details will be put into ISS file.
BBB	Abbreviation of MO.
999999	MO's code, according to code table which was registered with Napas
X	The order number of payment session (in a day). Noted : MO needs to take the file with the highest number of X, Normally X =1 daily, unless something goes wrong then X= 2 or more, with these values NAPAS could communicate with MO.
TC	The report file of successful transactions in a session is created by NAPAS and sent to MO.
SL	The dispute transactions report file is created by MO and sent to Napas
XL	The transaction detail file which contains the processed transactions in a session is created by Napas and sent to MO.
<ServiceCode>	Each transaction detail file includes transactions corresponding to their services, they have following service codes. 1. SWC: includes transactions at the ATM / POS (withdrawals, payments, etc.); transactions of services that connect to the APN; withdrawal transactions using Payment Code; inter-bank fund transfer 24/7 service (IBFT, CBFT); international cash disbursement by card services 2. PC: includes transactions to create Payment Code. 3. Ecom: includes online switching transactions, Tokenization transactions. 4. UPI: includes transactions of the services which connect to UPI.

4.2 Reconciliation file list

The list of reconciliation files are used between NAPAS and MO:

File	Description	File Name format
TC01	The detail file contains the successful transactions in a session is created by Napas and sent to the Issuer	MMDDYY_ISS_BBB_999999_X_TC_<ServiceCode>.dat
TC02	The detail file contains the successful transactions in a session is created by Napas and sent to the Acquirer	MMDDYY_ACQ_BBB_999999_X_TC_<ServiceCode>.dat
TC03	The file contains the successful transactions in a session is created by Napas and sent to the Beneficiary organizations	MMDDYY_BNB_BBB_999999_X_TC_<ServiceCode>.dat
SL01	The file contains the dispute transactions in a session is created by the Issuer and sent to NAPAS	MMDDYY_ISS_BBB_999999_X_SL_<ServiceCode>.dat
SL02	The file contains the dispute transactions in a session. This file created by the Acquirer and sent to NAPAS	MMDDYY_ACQ_BBB_999999_X_SL_<ServiceCode>.dat
SL03	The file contains the dispute transactions in a session is created by the Beneficiary organizations and sent to NAPAS	MMDDYY_BNB_BBB_999999_X_SL_<ServiceCode>.dat
XL01	The file contains the processed transactions in a session is created by Napas and sent to the Issuer	MMDDYY_ISS_BBB_999999_X_XL_<ServiceCode>.dat
XL02	The file contains the processed transactions in a session is created by Napas and sent to the Acquirer	MMDDYY_ACQ_BBB_999999_X_XL_<ServiceCode>.dat

XL03	The file contains the processed transactions in a session created by Napas and sent to the Beneficiary organizations	MMDDYY_BNB_BBB_999999_X_XL_<ServiceCode>.dat
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Note: In case, the MO does not find any dispute transactions, the MO sends these files (SL01, SL02 and SL03) to NAPAS to inform that there is not any dispute at MO. But these files (SL01, SL02 and SL03) only contain the header line, trailer line and do not contain the detail lines.

4.3 Reconciliation file format

The general rules in filling data between NAPAS and MO through reconciliation process:

- For filling data, always align right
- If the type is number, padding left with zero
- If the type is character, padding left with space

4.3.1 Successful transaction file and processed transaction file in a session

The detail file contains the successful transactions in a session created by Napas and sent to the MO: MMDDYY_ZZZ_BBB_999999_X_TC_<ServiceCode>.dat

The detail file contains the processed transactions in a session created by Napas and sent to the MO: MMDDYY_ZZZ_BBB_999999_X_XL_<ServiceCode>.dat

The file format consists of 03 parts:

- Header line (Header): describes the file header information.
- Detail lines (Detail): each transaction corresponds to a line in this file. Each transaction line ends with a checksum field to verify the integrity of the data.
- Trailer line (Trailer): contains information such as the number of transactions in the file, the file creator, creation date of file, checksum value of file.

After receiving the file, the MO has to check each line in the file and the line checksum value to verify the integrity of the line, and check the file checksum value to verify the integrity of the file.

4.3.1.1 Header line format (Header)

No	Field Name	Type	Length	Value	Description
1.	Record type	Character	2	HR	Record type HR = Header Record
2.	Field identification code	Character	5	[REV]	Identification code of "Receiver bank" field (Receiver – REV)
	Receiver Bank code	Character	8		The receiver bank code must be same with the bank code in filename The code is right indented and padded space in the left if there is less than 08 characters
3.	Field identification code	Character	6	[DATE]	Identification code of "Transaction date (DATE)" field
	Transaction date	Number	8		Transaction date of last transaction in the reconciliation file In ddMMyyyy format: dd: Date MM: Month yyyy: Year

4.3.1.2 Detail line format (Details)

No	Field Name	Type	Length	Value	Description
1.	Record type	Character	2	DR	Record type DR = Detail Record
2.	Field identification	Character	5	[MTI]	Identification code of MTI field

	code				
	Message Type Indicator (MTI)	Number	4		Message Type Indicator
3.	Field identification code	Character	4	[F2]	Identification code of F2 field
	Primary Account Number (F2)	Character	19		Card number is right indented and padding space in the left if there is less than 19 characters
4.	Field identification code	Character	4	[F3]	Identification code of F3 field
	Processing Code (F3)	Number	6		Transaction processing code
5.	Field identification code	Character	5	[SVC]	Identification code of Service Code (SVC) field
	Service Code (F62)	Character	10		Service Code – specified by NAPAS
6.	Field identification code	Character	5	[TCC]	Identification code of Transaction Channel Code (TCC) field (F60)
	Transaction Channel Code	Character	2		Identification code of Transaction Channel field (F60)
7.	Field identification code	Character	4	[F4]	Identification code of F4 field
	Transaction Amount (F4)	Number	12		Transaction amount is converted in VND if the transaction performed oversea. It has 02 decimal digits, padding left with zero if it's value is less than 12

					digits
8.	Field identification code	Character	5	[RTA]	Identification code of of Real Transaction Amount field
	Real Transaction Amount	Number	12		Transaction amount is converted in VND if the transaction performed oversea. Transaction Amount in VND currency, with 02 decimal digits, padding left with zero if it's value is less than 12 digits. If transaction is successful and is not reversed, then this amount is equal to the transaction amount (F4). If transaction is successful and is reversed, then this amount is zero. If transaction is successful and is reversed partialy, then this amount is less than the transaction amount (F4).
9.	Field identification code	Character	5	[F49]	Identification code of F49 field.
	Currency code (F49)	Number	3		The currency code with 03 numbers, padding left with zero if it's value is less than 03 digits.
10.	Field identification code	Character	4	[F5]	Identification code of F5 field
	Settlement Amount (F5)	Number	12		02 decimal digits, padding left with zero if it's value is less than 12

					digits
11.	Field identification code	Character	5	[F50]	Identification code of F50 field
	Settlement currency code (F50)	Number	3		The currency code with 03 numbers, padding left with zero if it's value is less than 03 digits
12.	Field identification code	Character	4	[F9]	Identification code of F9 field
	Settlement exchange rate (F9)	Number	8		This field format : nXXXXXXX n : number of decimal digits. XXXXXXX : exchange rate Fill zero if there is no value.
13.	Field identification code	Character	4	[F6]	Identification code of F6 field
	Billing Amount (F6)	Number	12		02 decimal digits, padding left with zero if it's value is less than 12 digits
14.	Field identification code	Character	5	[RCA]	Identification code of Real Cardholder Amount field
	Real Cardholder Amount	Number	12		02 decimal digits, padding left with zero if it's value is less than 12 digits. If transaction is successful and is not reversed, then this amount is equal to the transaction amount. If transaction is successful and is

					reversed, then this amount is zero. If transaction is successful and is reversed partially, then this amount is less than the transaction amount.
15.	Field identification code	Character	5	[F51]	Identification code of F51 field
	Cardholder Billing currency code (F51)	Number	3		The currency code with 03 numbers, padding left with zero if it's value is less than 03 digits
16.	Field identification code	Character	5	[F10]	Identification code of F10 field
	Cardholder billing conversion rate (F10)	Number	8		This field format : nXXXXXXX n : number of decimal digits. XXXXXXX : exchange rate Fill with zero if there is no value.
17.	Field identification code	Character	5	[F11]	Identification code of F11 field
	System Trace Audit Number (F11)	Number	6		Padding left with zero if there is less than 06 digits
18.	Field identification code	Character	5	[F12]	Identification code of F12 field
	Local transaction time (F12)	Number	6		In HHmmss format HH : hour (24-hours format) mm : minute ss : second

19.	Field identification code	Character	5	[F13]	Identification code of F13 field
	Local transaction date (F13)	Number	4		In MMdd format MM : month dd : day of month
20.	Field identification code	Character	5	[F15]	Identification code of F15 field
	Settlement date (F15)	Number	4		In MMdd format MM : month dd : day of month
21.	Field identification code	Character	5	[F18]	Identification code of F18 field
	Merchant category code (F18)	Number	4		Merchant type
22.	Field identification code	Character	5	[F22]	Identification code of F22 field
	Point-of-service Entry Mode (F22)	Number	3		The entry mode at Point-of-service The code identifies how the PAN and PIN is inputted at the Point-of-service
23.	Field identification code	Character	5	[F25]	Identification code of F25 field
	Point-of-service Condition Code (F25)	Number	2		The condition code at Point-of-service

24.	Field identification code	Character	5	[F41]	Identification code of F41 field For ECOM service : filled with merchant ID (numeric), padding left with zero if it's value is less than 8 characters
	Card acceptor terminal identification (F41)	Character	8		A unique code to identify a card acceptor in acquirer bank system
25.	Field identification code	Character	5	[ACQ]	Identification code of Acquiring institution code field (F32)
	Acquiring institution code (F32)	Character	8		Acquirer identification code is registered in Napas Padding left with zero if it's value is less than 08 digits
26.	Field identification code	Character	5	[ISS]	Issuer identification code (Issuer Identify)
	Issuing identification code	Character	8		Issuer identification code was registered in Napas It is right indented, padding left with zero if it's value is less than 08 digits.
27.	Field identification code	Character	5	[MID]	Identification code of card acceptor identification code field (F42)
	Card acceptor identification code (F42)	Character	15		The unique code identifies card acceptor identification code in POS transaction For Ecom service, this field record card acceptor identification

					code (characters), if it's value is more than 15 numbers, only take the first 15 numbers. For IBFT service, this field record Payment Intermediary identification code It is right indented, padding left with zero if it's value is less than 15 digits
28.	Field identification code	Character	5	[BNB]	Identification code of F100 (Beneficiary Identify) field
	Beneficiary identification code (F100)	Character	8		For IBFT, padding left with zero if it's value is less than 08 digits
29.	Field identification code	Character	6	[F102]	Identification code of F102 field
	Source Account Number(F102)	Character	28		Padding left with space if it's value is less than 28 digits
30.	Field identification code	Character	6	[F103]	Identification code of F103 field
	Destination Card/Account number (F103)	Character	28		Padding left with space if it's value is less than 28 digits
31.	Field identification code	Character	10	[SVFISS NP]	Identification code of ISS service fee for NAPAS field
	ISS Fee (service fee of ISS for NAPAS)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits

					Fill with zero if there is no value
32.	Field identification code	Character	11	[IRFISS ACQ]	Identification code of ISS interchange fee for ACQ field
	ISS Fee (interchange fee of ISS for ACQ)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
33.	Field identification code	Character	11	[IRFISS BNB]	Identification code of ISS interchange fee for BNB field
	ISS Fee (interchange fee of ISS for BNB)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
34.	Field identification code	Character	10	[SVFAC QNP]	Identification code of ACQ service fee for NAPAS field
	ACQ Fee (service fee of ACQ for NAPAS)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
35.	Field identification code	Character	11	[IRFACQ ISS]	Identification code of ACQ interchange fee for ISS field
	ACQ Fee (interchange fee of ACQ for ISS)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value

36.	Field identification code	Character	11	[IRFACQ BNB]	Identification code of ACQ interchange fee for BNB field
	ACQ Fee (interchange fee of ACQ for BNB)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
37.	Field identification code	Character	10	[SVFBN BNP]	Identification code of BNB service fee for NAPAS field
	BNB Fee (service fee of BNB for NAPAS)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
38.	Field identification code	Character	11	[IRFBNB ISS]	Identification code of BNB interchange fee for ISS field
	BNB Fee (interchange fee of BNB for ISS)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
39.	Field identification code	Character	11	[IRFBNB ACQ]	Identification code of BNB interchange fee for ACQ field
	BNB Fee (interchange fee of BNB for ACQ)	Number	12		Padding left with zero if it's value is less than 12 digits 02 decimal digits, Fill with zero if there is no value
40.	Field identification	Character	5	[F37]	Identification code of F37 field.

	code				For ECOM service : filled with transaction code.
	Retrieval Reference number (F37)	Character	12		Transaction Retrieval Reference number Padding left with space if it's value is less than 12 digits
41.	Field identification code	Character	5	[F38]	Identification code of F38 field
	Authorization Identification Response (F38)	Character	6		Authorization Number is responded by Issuer Fill the space if there is no value
42.	Field identification code	Character	5	[TRN]	Identification code of F63 field (Transaction Reference Number – TRN)
	Transaction Reference Number – TRN (F63)	Character	16		The transaction identification number is generated by Napas when the transaction is processed
43.	Field identification code	Character	5	[RRC]	Identification code of Reconciliation response code field (RRC)
	Reconciliation Response Code	Number	4		Reconciliation Response Code is specified in the settlement process for MO Padding left with zero if it's value is less than 4 digits.
44.	Field identification code	Character	6	[RSV1]	Reserve information field 1
	Reserve information 1	Character	100		- For ECOM service : Noted that this field to

					add and fill with full values using with merchants in ECOM service. + From 1-32 (including 32 characters) : filled with merchant code (but from Field 48 of online message), padding right with space if it is less than 32 characters. + From 33-52 (including 20 characters) : filled with transaction code (but from Field 48 of online message), padding right with space if it is less than 20 characters. + From 53-100 : not used, filled with spaces. - For the other services : filled with spaces
45.	Field identification code	Character	6	[RSV2]	Reserve information field 2
	Reserve information 2	Character	100		At present, it is not used. Reserved for future, filled with space characters
46.	Field identification code	Character	6	[RSV3]	Reserve information field 3
	Reserve information 3	Character	100		At present, it is not used. Reserved for future, filled with space characters
47.	Field identification	Character	5	[CSR]	Identification code of Checksum field (CSR)

	code				
	Checksum value	Character	32		To verify the integrity of data line Checksum is calculated using MD5 hash

4.3.1.3 Trailer line (Trailer)

No	Field Name	Type	Length	Value	Description
1.	Record type	Character	2	TR	Record type TR = Trailer Record
2.	Field identification code	Character	5	[NOT]	Identification code of NOT (Number Of Transactions) Field
	Number Of Transactions	Number	9		The number of transaction lines in file
3.	Field identification code	Character	5	[CRE]	Identification code of Creator field
	File creator	Character	20		The user identification of the system which exports the file Padding left with zero if it's value is less than 20 digits
4.	Field identification code	Character	6	[TIME]	Identification code of Time field (TIME)
	Creation time	Number	6		The creation time on the system that exports the file in HHmmss format HH : hour (24-hours format) mm : minute

					ss : second
5.	Field identification code	Character	6	[DATE]	Identification code of file creation date field (DATE)
	Creation date	Number	8		The creation date is calculated same with the system date that exports the file In ddMMyyyy format MM : month dd : day of month yyyy : year
6.	Field identification code	Character	5	[CSF]	Identification code of Checksum field (CSF)
	Checksum value	Number	32		To verify the integrity of the trailer line, to check a summary of line on creation file Checksum is calculated using MD5 hash

4.3.2 Dispute transaction file

The detail file contains the dispute transactions in a session created by the MO and sent to Napas: **MMDDYY_ZZZ_BBB_999999_X_SL_<ServiceCode>.dat**

The file format consists of 03 parts:

- Header line: describes the file header information.
- Detail lines: each transaction corresponds to a line in this file. Each transaction line ends with a checksum field to verify the integrity of the data.
- Trailer end: contains information such as the number of transactions in the file, the creator file, file creation date, file checksum value.

After receiving the file, the Napas has to check each line in the file and the line checksum value to verify the integrity of the line, and check the file checksum value to verify the integrity of the file.

4.3.2.1 Header line format (Header)

No	Field Name	Type	Length	Value	Description
1.	Record type	Character	2	HR	Record type HR = Header Record
2.	Field identification code	Character	5	[REV]	Identification code of "Receiver bank" field (Receiver – REV)
	Receiver Bank code	Character	8		The receiver bank code must be same with the bank code in filename The code is right indented and padded space in the left if there is less than 08 characters
3.	Field identification code	Character	6	[DATE]	Identification code of "Transaction date (DATE)" field
	Transaction date	Number	8		Transaction date of last transaction in the reconciliation file In ddMMyyyy format: dd: Date MM: Month yyyy: Year

4.3.2.2 Detail line format (Details)

No	Field Name	Type	Length	Value	Description
1.	Record type	Character	2	DR	Record type DR = Detail Record
2.	Field identification	Character	5	[MTI]	Identification code of MTI field

	code				
	Message Type Indicator (MTI)	Number	4		Message Type Indicator
3.	Field identification code	Character	4	[F2]	Identification code of F2 field
	Primary Account Number (F2)	Character	19		Card number is right indented and padding space in the left if there is less than 19 characters
4.	Field identification code	Character	4	[F3]	Identification code of F3 field
	Processing Code (F3)	Number	6		Transaction processing code
5.	Field identification code	Character	5	[SVC]	Identification code of Service Code (SVC) field
	Service Code (F62)	Character	10		Service Code – specified by NAPAS
6.	Field identification code	Character	5	[TCC]	Identification code of Transaction Channel Code (TCC) field (F60)
	Transaction Channel Code	Character	2		Identification code of Transaction Channel field (F60)
7.	Field identification code	Character	4	[F4]	Identification code of F4 field
	Transaction Amount (F4)	Number	12		Transaction amount is converted in VND if the transaction performed oversea. It has 02 decimal digits, padding left with zero if it's value is less than 12

					digits
8.	Field identification code	Character	5	[RTA]	Identification code of of Real Transaction Amount field
	Real Transaction Amount	Number	12		Transaction amount is converted in VND if the transaction performed oversea. Transaction Amount in VND currency, with 02 decimal digits, padding left with zero if it's value is less than 12 digits. If transaction is successful and is not reversed, then this amount is equal to the transaction amount (F4). If transaction is successful and is reversed, then this amount is zero. If transaction is successful and is reversed partialy, then this amount is less than the transaction amount (F4).
9.	Field identification code	Character	5	[F49]	Identification code of F49 field.
	Currency code (F49)	Number	3		The currency code with 03 numbers, padding left with zero if it's value is less than 03 digits.
10.	Field identification code	Character	4	[F5]	Identification code of F5 field
	Settlement Amount (F5)	Number	12		02 decimal digits, padding left with zero if it's value is less than 12 digits

11.	Field identification code	Character	5	[F50]	Identification code of F50 field
	Settlement currency code (F50)	Number	3		The currency code with 03 numbers, padding left with zero if it's value is less than 03 digits
12.	Field identification code	Character	4	[F9]	Identification code of F9 field
	Settlement conversion rate (F9)	Number	8	F9	This field format : nXXXXXXX n : number of decimal digits. XXXXXXX : exchange rate Fill zero if there is no value.
13.	Field identification code	Character	4	[F6]	Identification code of F6 field
	Cardholder Billing Amount (F6)	Number	12		02 decimal digits, padding left with zero if it's value is less than 12 digits
14.	Field identification code	Character	5	[RCA]	Identification code of Real Cardholder Amount field
	Real Cardholder Amount	Number	12		02 decimal digits, padding left with zero if it's value is less than 12 digits. If transaction is successful and is not reversed, then this amount is equal to the transaction amount. If transaction is successful and is reversed, then this

					amount is zero. If transaction is successful and is reversed partialy, then this amount is less than the transaction amount.
15.	Field identification code	Character	5	[F51]	Identification code of F51 field
	Cardholder Billing currency code (F51)	Number	3		The currency code with 03 numbers, padding left with zero if it's value is less than 03 digits
16.	Field identification code	Character	5	[F10]	Identification code of F10 field
	Cardholder Billing Conversion Rate (F10)	Number	8		This field format : nXXXXXXXX n : number of decimal digits. XXXXXXXX : exchange rate Fill with zero if there is no value.
17.	Field identification code	Character	5	[F11]	Identification code of F11 field
	System Trace Audit Number (F11)	Number	6		Padding left with zero if there is less than 06 digits
18.	Field identification code	Character	5	[F12]	Identification code of F12 field
	Local transaction time (F12)	Number	6		In HHmmss format HH : hour (24-hours format) mm : minute ss : second

19.	Field identification code	Character	5	[F13]	Identification code of F13 field
	Local transaction date (F13)	Number	4		In MMdd format MM : month dd : day of month
20.	Field identification code	Character	5	[F15]	Identification code of F15 field
	Settlement date (F15)	Number	4		In MMdd format MM : month dd : day of month
21.	Field identification code	Character	5	[F18]	Identification code of F18 field
	Merchant Category Code (F18)	Number	4		Merchant type
22.	Field identification code	Character	5	[F22]	Identification code of F22 field
	Point-of-service Entry Mode (F22)	Number	3		The entry mode at Point-of-service The code identifies how the PAN and PIN is inputted at the Point-of-service
23.	Field identification code	Character	5	[F25]	Identification code of F25 field
	Point-of-service Condition Code (F25)	Number	2		The condition code at Point-of-service

24.	Field identification code	Character	5	[F41]	Identification code of F41 field
	Card Acceptor Terminal Identification (F41)	Character	8		A unique code to identify a card acceptor in acquirer bank system. For ECOM service : filled with merchant ID (characters with numeric format), padding left with zero if it's value is less than 8 characters
25.	Field identification code	Character	5	[ACQ]	Identification code of Acquiring institution code field (F32)
	Acquiring Institution Code (F32)	Character	8		Acquirer identification code is registered in Napas Padding left with zero if it's value is less than 08 digits
26.	Field identification code	Character	5	[ISS]	Issuer identification code (Issuer Identify)
	Issuing Identification Code	Character	8		Issuer identification code was registered in Napas It is right indented, padding left with zero if it's value is less than 08 digits.
27.	Field identification code	Character	5	[MID]	Identification code of card acceptor identification code field (F42)
	Card Acceptor Identification Code	Character	15		The unique code identifies a card acceptor in POS transactions. For Ecom service, this field record merchant identification code

					(character), if it's value is more than 15 numbers, only take the first 15 numbers. For IBFT service, this field record Payment Intermediary code It is right indented, padding left with zero if it's value is less than 15 digits
28.	Field identification code	Character	5	[BNB]	Identification code of F100 (Beneficiary Identify) field
	Beneficiary Identification Code (F100)	Character	8		For IBFT, padding left with zero if it's value is less than 08 digits
29.	Field identification code	Character	6	[F102]	Identification code of F102 field
	Source Account Number (F102)	Character	28		Padding left with space if it's value is less than 28 digits
30.	Field identification code	Character	6	[F103]	Identification code of F103 field
	Destination Card/Account number (F103)	Character	28		Padding left with space if it's value is less than 28 digits
31.	Field identification code	Character	5	[F37]	Identification code of F37 field. For ECOM service : filled with transaction code.
	Retrieval Reference Number (F37)	Character	12		Reference number transaction for reconciliation

					Padding left with space if it has less than 12 characters
32.	Field identification code	Character	5	[F38]	Identification code of F38 field
	Authorization Identification Response (F38)	Character	6		Authorization Number is responded from Issuer. Fill space if there is no value
33.	Field identification code	Character	5	[TRN]	Identification code of F63 field (Transaction Reference Number – TRN)
	Transaction Reference Number – TRN (F63)	Character	16		The transaction identification number is generated by Napas when the transaction is processed
34.	Field identification code	Character	5	[RRC]	Identification code of Reconciliation response code field (RRC)
	Reconciliation Response Code	Number	4		Reconciliation Response Code is specified in the settlement process for MO. Padding left with zero if it's value is less than 4 digits.
35.	Field identification code	Character	6	[RSV1]	Reserve information field 1
	Reserve information 1	Character	100		- For ECOM service : Noted that this field to add and fill with full values using with merchants in ECOM service. + From 1-32 (including 32 characters) : filled

					with merchant code (but from Field 48 of online message), padding right with space if it is less than 32 characters. + From 33-52 (including 20 characters) : filled with transaction code (but from Field 48 of online message), padding right with space if it is less than 20 characters. + From 53-100 : not used, filled with spaces. - For the other services : filled with spaces
36.	Field identification code	Character	6	[RSV2]	Reserve information field 2
	Reserve information 2	Character	100		At present, it is not used. Reserved for future, filled with space characters
37.	Field identification code	Character	6	[RSV3]	Reserve information field 3
	Reserve information 3	Character	100		At present, it is not used. Reserved for future, filled with space characters
38.	Field identification code	Character	5	[CSR]	Identification code of Checksum field (CSR)
	Checksum value	Character	32		To verify the integrity of data line Checksum is calculated using MD5 hash

4.3.2.3 Trailer line format (Trailer)

No	Field Name	Type	Length	Value	Description
1.	Record type	Character	2	TR	Record type TR = Trailer Record
2.	Field identification code	Character	5	[NOT]	Identification code of NOT (Number Of Transactions) Field
	Number Of Transactions	Number	9		The number of transaction lines in file
3.	Field identification code	Character	5	[CRE]	Identification code of Creator field
	File creator	Character	20		The user identification of the system which exports the file Padding left with zero if it's value is less than 20 digits
4.	Field identification code	Character	6	[TIME]	Identification code of Time field (TIME)
	Creation time	Number	6		The creation time on the system that exports the file in HHmmss format HH : hour (24-hours format) mm : minute ss : second
5.	Field identification code	Character	6	[DATE]	Identification code of file creation date field (DATE)
	Creation date	Number	8		The creation date is calculated same with the system date that exports the file

					In ddMMyyyy format MM : month dd : day of month yyyy : year
6.	Field identification code	Character	5	[CSF]	Identification code of Checksum Field (CSF)
	Checksum value	Number	32		To verify the integrity of the trailer line, to check a summary of line on creation file Checksum is calculated using MD5 hash

5 Effective document

The NAPAS Technical Specification for switching service takes effect from **01st January, 2017.**

6 Document management

Internal reference document:

No.	Document Name
1	BANKNETVN Technical Specification Document (version 1.6.1).
2	SMARTLINK Technical Specification Document (version 2.0).
3	Regulation for Member Organization Participation on the SMARTLINK System.
4	Regulation for Member Organization Participation on the BANKNETVN System.
5	Regulation for Business Settlement and Reconciliation Process on the BANKNETVN System (released at 28th February, 2013).

External reference document:

No.	Document Name
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1	ISO 8583-1987 Technical Specification Document.
2	Technical Specification Documents of VISA Organization.
3	Technical Specification Documents of MASTERCARD Organization.
4	Technical Specification Documents of UPI Organization.